EduDocAI System Architecture Whitepaper (Page 1)

[DEMONSTRATION SYNTHETIC DATASET - RESEARCH ARCHITECTURE DOCUMENT]
EduDocAI System Architecture & Engineering Specifications

1. Pipeline Overview
EduDocAI uses a hybrid retrieval-augmented generation (RAG) architecture.
Document ingestion processes PDF and text streams in-memory using BytesIO to prevent OS file locks.
Text is extracted page-by-page using PyMuPDF, preserving document ID, source filename, and page numbers.

2. Chunking Configuration
The default chunk size is 400 tokens with an overlap of 60 tokens.
Sentence-boundary segmentation is performed using regular expression boundary heuristics to avoid truncating propositions.

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3. Embedding and Vector Indexing
Dense retrieval uses the sentence-transformers all-MiniLM-L6-v2 model, producing 384-dimensional dense vectors. Vectors are normalized using L2 normalization to allow inner-product calculation to represent exact cosine similarity. FAISS IndexFlatIP is employed for vector indexing.

4. Sparse TF-IDF Baseline
TF-IDF vectors are constructed using sublinear term frequency scaling and bi-gram ranges (1, 2).
Reciprocal Rank Fusion (RRF) with constant k=60 is available for hybrid reranking.